Chaoan Li

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Education

Texas A&M University — College Station, TX
Ph.D. Student, Industrial Engineering, Data Science Track · GPA: 4.0/4.0 — *Aug. 2024 – May 2028 (expected)*

Beijing Normal University — Beijing, China
M.S., Mathematics · GPA: 3.7/4.0 — *Sep. 2021 – Jul. 2024*

University of Science and Technology Beijing — Beijing, China
B.S., Applied Mathematics · Minor: Financial Engineering · GPA: 3.94/4.0 — *Sep. 2017 – Jul. 2021*

Summary

Researcher developing statistical methods for LLM evaluation, model validation, and uncertainty quantification. Combines mathematical research with reusable Python tools and deployed applications, including an LLM grading workflow used during a teaching assistantship.

Technical Skills

Machine Learning: PyTorch; LLMs (evaluation, watermarking, text generation, privacy-risk analysis); Transformer, Mamba, CNN, RNN, LSTM; decision trees, SVM, kernel methods, nonlinear modeling
Mathematics & Statistics: High-dimensional statistics, uncertainty quantification, Monte Carlo simulation, functional analysis, variational methods
Programming: Python, R, SQL, C++, C#, MATLAB
Libraries & Tools: Git, scikit-learn, NumPy, pandas, SciPy, Jupyter, Linux, Bash

Research Experience

Texas A&M University — College Station, TX
Research Assistant (Advisor: Prof. Rui Tuo) — *Aug. 2024 – Present*

LLM Privacy: Detecting Watermark Reuse Without Secret Keys
- Developed Python/PyTorch tests to detect shared watermark keys within text groups, **outperforming four text-similarity baselines** in short-context Gemma and Qwen experiments. Diagnosed topic confounding and quantified how context design and key rotation change detection power.

Model Validation: Detecting and Localizing Simulator Errors
- Detected and localized a shear-layer discrepancy **missed by two baseline tests** using FMMT, a frequency-domain method. Built a reusable Python toolkit and extended inference to account for simulator parameters estimated from the same data.

Nonlinear Model Calibration and Uncertainty Quantification
- Unified three simulator-calibration approaches in a modular Python framework for parameter estimation and uncertainty quantification. Demonstrated how fitting criteria change parameters and confidence intervals in a cardiac ion-channel model.

Beijing Normal University — Beijing, China
Research Assistant (Advisor: Prof. Dachun Yang) — *Jan. 2022 – Jul. 2024*

Geometry and Characterization of Function Spaces
- Published **two first-author peer-reviewed papers** on mathematical representations of complex functions, establishing decomposition, duality, and Fourier characterizations.

Selected Projects

EasyGrader: LLM-Assisted Homework Grading | Live App | Code
- Automated rubric-based scoring and feedback with a deployed React/TypeScript and Express application **used during a teaching assistantship**; supported instructor-defined assignments and grading policies.
- Integrated four LLM providers with concurrent batch processing and retries; accepted PDF, Jupyter notebook, and code submissions and exported results to CSV for instructor review.

Interactive Option Analytics Application | Live App | Code
- Integrated Yahoo Finance option-chain data and U.S. Treasury yields into a deployed React/TypeScript application with Vercel serverless functions, enabling configurable option pricing and sensitivity analysis.
- Implemented Black–Scholes–Merton pricing, an American-option binomial tree, and Newton–Raphson implied-volatility estimation; provided five option Greeks in an interactive English/Chinese interface.

Selected Publications (2 published, 2 under review)

C. Li and R. Tuo. A Unified Frequentist Framework for Computer Model Calibration. **IISE Transactions** (under review), 2026.
C. Li, X. Zhang, and R. Tuo. Statistical Validation of Computer Models: Global and Subdomain Hypothesis Testing. **Technometrics** (under review), 2026.
C. Li, X. Yan, and D. Yang. Fourier Transform of Anisotropic Hardy Spaces Associated with Ball Quasi-Banach Function Spaces and Its Applications to Hardy–Littlewood Inequalities. **Acta Mathematicae Applicatae Sinica, English Series**, 2024.
C. Li, X. Yan, and D. Yang. Anisotropic Ball Campanato-Type Function Spaces and Their Applications. **Analysis and Mathematical Physics**, 13(3), Article 50, 2023.

Honors & Awards

NeurIPS Financial Assistance Award, full conference and workshop registration — 2026
Selected Oral Presenter, CRSE Workshop, The Wharton School, University of Pennsylvania — 2026
National Scholarship, China — 2020
Meritorious Winner, U.S. Mathematical Modeling Competition — 2019
First Prize, Beijing Undergraduate Mathematical Modeling Competition — 2019